1 Forward, Futures & Hedging

Formulas

$$\sum_{i=0}^{n-1} ar^i = \frac{a(1-r^n)}{1-r}$$

For given rate $r\%$ over time period $t \in [0, T]$, with annual period P, discount factor $d_{0,T}$ is given

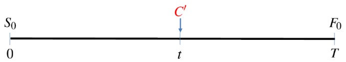
$$d_{0,T} = \left(1 + r\% \times \frac{T}{P}\right)^{-1} < 1$$

No Arbitrage Argument

To prove $A = B$, show strategies for $A \neq B$. If $A > B$, short on A, long on B, and vice versa.

1.1.2 Asset w/ Predictable Income

$$F_0 = (S_0 - C)/d_{0,T}, C = C' d_{0,t}$$

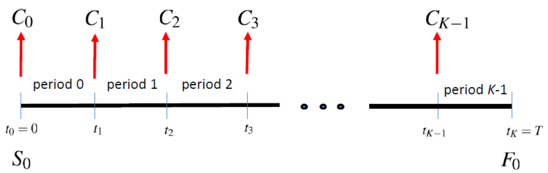


1.1.3 Asset w/ Continuous Dividend Yield

Annualized continuous dividend yield of q

$$F_0 = S_0 \exp((r-q)T)$$

1.1.4 Asset w/ Storage Costs

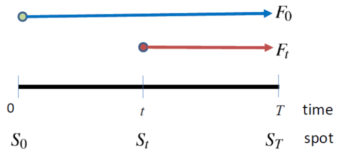


$$d_{0,K} F_0 = S_0 + \sum_{j=0}^{K-1} d_{0,j} C_j$$

1.1.5 Convenience Yield

$$d_{0,K} F_0 = S_0 + \sum_{j=0}^{K-1} d_{0,j} (C_j - y)$$

1.2 Value of Forward



Value of **long** position:

$$f_t = (F_t - F_0) d_{t,T}$$

1.3.1 Marked-to-Market Mechanism

1. Investor deposits intial margin
2. Account is marked to market each day on gain/loss of futures position
3. Margin account balance flutuates day to day
4. If balance drops below threshold, investor required to top up with cash

1.3.2 Futures-Forwards Equivalence

If interest rates are known to follow expectation dynamics, then theoretical futures and forward prices of corresponding contracts are identical.

1.4.1 Short Hedge

To **lock in** a selling price, applicable when

- already owns asset and expects to sell in futures
- does not own asset but will in future, and sell later afterward

Net cash inflow of $F_0 = S_T + (F_0 - S_T)$ is locked in.
With basis risk,

$$F_1 + b_2 = S_2 + (F_1 - F_2)$$

1.4.2 Long Hedge

To **lock in** a purchase price, applicable when hedger has to purchase asset in the future.
Net cash outflow of $F_0 = S_T - (S_T - F_0)$ is locked in.
With basis risk,

$$F_1 + b_2 = S_2 - (F_2 - F_1)$$

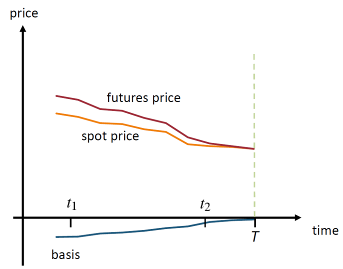
1.4.3 Basis Risk

Perfect hedges often not possible in reality.

1. unavailability of futures contract matching asset obligation dates
2. quantity of asset obligated may not be integer multiple of contract size
3. lack of liquidity in futures market
4. terms of delivery may not coincide with obligation
5. unavailability of futures contracts with underlying asset exactly matching asset hedged

Define basis b_i at time t_i, with spot price S_i and futures price F_i as

$$b_i = S_i - F_i$$



Effective price can be written as

$$F_1 + b_2 = S_1 - (b_1 - b_2)$$

Price at time t_1, offset by diff btw **initial basis** b_1 and **cover basis** b_2.
Should choose futures contract with maturity close, but later than expiration of hedge.

1.4.4 Cross Hedging

Futures and spot price of proxy asset at time t_i denoted by F_i^* and S_i^* respectively. Effective price achieved is

$$S_2 + (F_1^* - F_2^*) = F_1^* + (S_2^* - F_2^*) + (S_2 - S_2^*)$$

2 Forwards & Futures in Practice

2.1.1 US Treasury Bills

- **Dollar price**: Typically quoted per \$100 of face value
- Often given as **discount yield** d

$$d = \frac{Par - P}{Par} \times \frac{360}{D}$$

where D is days to maturity.

- Dollar Price is then

$$P = Par \times \left(1 - d \times \frac{D}{360}\right)$$

- True yield is the **coupon equivalent yield** (CEY) or **investment rate**

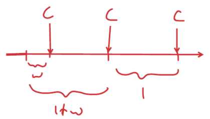
1. T-bills not more than half-year to maturity

$$i = \frac{100 - P}{P} \times \frac{y}{t}$$

2. T-bills more than half-year to maturity. Root of

$$\left(\frac{t}{2y} - 0.25\right) i^2 + \frac{t}{y} i + \frac{P - 100}{P} = 0$$

2.1.2 Accrued Interest



Bond price a.k.a. **full, dirty, or invoice price**

$$P = \frac{F}{\left(1 + \frac{r}{m}\right)^{w+n-1}} + \sum_{i=0}^{n-1} \frac{C}{\left(1 + \frac{r}{m}\right)^{w+i}}$$

Quoted prices a.k.a. **clean or flat price** does not include accrued interest.

$$DPrice = CPrice + AI$$

2.2.1 Repurchase Agreements

- Repurchase agreements = **repo** = **collateralized loan**.
- Underlying security held as **collateral** by counter-party.
- Counter-party entered **reverse repo**.
- Rate involved is **repo rate**.
- **Simple interest** rates. actual/360 in US.

1. At time $t = 0$, trader finances purchase of security worth S_0.
2. Enter T-day repo with Mr. Y.
3. Mr. Y demands haircut of h% and repo rate r.
4. Mr. Y lends $S_0(1 - h\%)$.
5. Trader has net outflow of $S_0 \times h\%$.
6. At time $t = T$, trader pays $F = S_0(1-h\%)(1+r\times T/360)$.
7. Trader receives security.
8. Equivalent to forward purchase of security at forward price F.
9. If security price increases to S_T then trader earns profit.
10. To carry out bet on decline using reverse repo, buy at S_0, sell at $S_0(1 - h\%)$, buy back at S_T.

2.2.2 SOFR

Secured Overnight Financing Rate (SOFR). Day count: actual/360. Compound SOFR Formula

$$r = \left[\prod_{i=1}^{T}\left(1 + \frac{r_i n_i}{Y}\right) - 1\right]\frac{Y}{d_c}$$

Simple SOFR Formula

$$r = \frac{1}{d_c}\sum_{i=1}^{T} r_i n_i$$

2.2.3 1-Month and 3-Month SOFR Futures

- 1-month SOFR Futures: \$5m loan for 1-month
- 3-month SOFR Futures: \$1m loan for 3-months
- SOFR Futures follow 30 day periods
- Quoted Futures price 1-month: $100 \times (1 - E(r_m))$, where r_m is the simple average SOFR for contract month
- Final Settlement price 1-month: $100 \times (1 - r_m)$.
- Quoted Futures price 3-months: $100 \times (1-$ expected compound average SOFR for contract period)
- Long position gains when expected SOFR falls (futures rises).
- Take up long position to hedge against rate falls.
- Long position is equivalent to agreeing to lend at rate

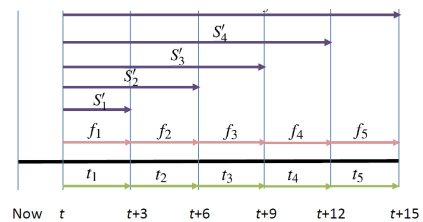
$$(100 - F_0)\%$$

- Short position gains when expected SOFR rises (futures falls).
- Take up Short position to hedge against rate rises.
- Short position is equivalent to agreeing to borrow at rate

$$(100 - F_0)\%$$

Changes to margin account

- 1-month: Decrease in futures price by 0.01 (rate rise by 1 bp) results in long position being decreased by \$41.67.
- 3-months: Decrease in futures price by 0.01 (rate rise by 1 bp) results in long position being decreased by \$25.



2.2.4 Treasury Bond Futures

Hypothetical 30-year 6% (semi-annual) coupon bond w par value of \$100,000.

$$P_{inv} = N \times F_T \times CF + AI$$

Conversion Factor (CF) is clean price of \$1 coupon bond with maturity equal to deliverable bond if priced to yield 6%.
With n remaining coupon payments,

$$P = \frac{1}{1.03^{w+n-1}} + \sum_{i=0}^{n-1} \frac{c}{1.03^{w+i}} - c(1-w)$$

$w \in \{0.5, 1\}$ is rounded to nearest quarter.

Cheapest-to-Deliver (CTD)

$$DPrice = CPrice + AI_t$$

Gross Basis

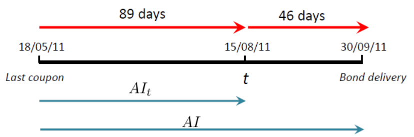
$$GB = CPrice - F_t \times CF$$

Net Basis

$$NB = DPrice \times (1 + r\frac{D}{360}) - (F_t \times CF + AI)$$

Implied Repo Rate (IRR)

$$IRR = \frac{(F_t \times CF + AI) - DPrice}{DPrice} \times \frac{360}{D}$$



2.2.5 Forward Rate Agreements (FRA)

E.g. FRA covering period one month and ending four months is 1×4 contract. Pricing FRAs, X is fixed rate (actual/360), y is actual ref rate (actual/360).

$$X = \left(\frac{d_{0,T}}{d_{0,T+m}} - 1\right) \times \frac{360}{m}$$

Value of FRA

$$v = [d_{t,T} - \left(1 + X \times \frac{m}{360}\right) \times d_{t,T+m}] \times N$$

2.3.1 Currency Prepaid Forward

$$F_0^P = x_0 d_{0,T}(r_u)$$

for interest rate r_u, current exchange rate x_0.

2.3.2 Currency Forward

$$F_0 = F_0^P / d_{0,T}(r_s)$$

Essentially the forward exchange rate.

2.3.3 Currency Futures

FX futures convention: **base-currency/terms-currency**.

3 Swaps

3.2.1 Swap Rate

$$N - d_{0,K}N = N(1 - d_{0,K})$$

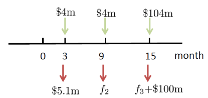
$$r = \frac{1 - d_{0,K}}{\sum_{i=1}^{K} \frac{t_i}{360} d_{0,i}}$$

3.2.2 Valuation of Swap

$$B_{fl} = (N + f_1)\exp(-r_1 t_1)$$

$$B_{fix} = N\exp(-r_n t_n) + \sum_{i=1}^{n} k\exp(-r_i t_i)$$

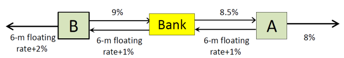
$$V = B_{fix} - B_{fl}$$



3.2.3 Arranging a Swap

A desires floating rate, B wants fixed rate.

	Fixed	Floating
A	**8%**	6-month floating rate + **1%**
B	11%	6-month floating rate + 2%



Outcome: A effectively borrows at 6-month floating rate+0.5% and B borrows at 10%.
Savings distribution: 0.5% for A, 1% for B, 0.5% for bank.

4 Options

4.1.1 European Vanilla Options

Holder of European call, payoff

$$\phi(T) = \max(0, S_T - X)$$

Profit

$$\phi(T) - c/d_{0,T}$$

Holder of European put, payoff

$$\phi(T) = \max(0, X - S_T)$$

Profit

$$\phi(T) - p/d_{0,T}$$

4.2.1 Options and Underliers

- Short Call, Long Asset (Buy-Write) (Covered Call)
- Long Put, Long Asset (Protective Put)

4.2.2 Spreads

Assume $X_i < X_j \Leftrightarrow i < j$

- Bull Spread - Long Call X_1, Short Call X_2

Total Profit
$+c_2^* - c_1^*$
$S_T - X_1 + c_2^* - c_1^*$
$X_2 - X_1 + c_2^* - c_1^*$

- Bear Spread - Short Put X_1, Long Put X_2

Total Profit
$X_2 - X_1 + p_1^* - p_2^*$
$X_2 - S_T + p_1^* - p_2^*$
$+p_1^* - p_2^*$

- Box Spread - Combine Bull + Bear Spread, $PV = (X_2 - X_1)d_{0,T}$
- Butterfly Spread - Long Call X_1, X_3, Short Call 2x X_2, $c_b = c_1 + c_3 - 2c_2$

Total Profit
$-c_b^*$
$S_T - X_1 - c_b^*$
$X_3 - S_T - c_b^*$
$-c_b^*$

4.2.3 Combination

- Straddle - Long Call, Long Put X
- Strip - Long Call, Long 2x Put X
- Strap - Long 2x Call, Long Put X
- Strangle - Long Call X_2, Long Put X_1

Total Payoff
$X_1 - S_T$
0
$S_T - X_2$

5 Value at Risk (VaR)

5.1.1 Definition

"We are c% confident (conf. lvl.) that we will not lose more than \$$V$ (VaR) in the next N days (risk horizon)."

$$P(L > VaR) \leq 1 - c$$

5.1.2 Non-parametric VaR

Let V_0, R be initial portfolio value and rate of return. μ, σ are mean and stdev.

$$V = V_0(1 + R)$$

Porfolio value V^* at rate of return R^*.

$$V^* = V_0(1 + R^*)$$

Relative VaR

$$VaR_r = E(V) - V^* = -V_0(R^* - \mu)$$

Absolute VaR

$$VaR_a = V_0 - V^* = -V_0 R^*$$

Short horizon, $\mu \approx 0, VaR_a \approx VaR_r$.

$$P\&L = V - V_0 = \Delta V = V_0 R$$

R is **risk factor**, V_0 is **dollar exposure to** R.
Given c, we find $(1-c)\times$ total obs. to the left, and interpolate between obs. to get $P\&L^*$.

5.1.3 Parametric VaR

$$R^* = \mu - \alpha\sigma$$

$$VaR_r = \alpha\sigma\sqrt{\Delta t}V_0$$

$$VaR_a = (\alpha\sigma\sqrt{\Delta t} - \mu)V_0$$

$$N - day VaR = 1 - day VaR \times \sqrt{N}$$

5.2 Valuation

5.2.1 Delta-Normal

$$\Delta V \approx \Delta_0 \Delta S = (\Delta_0 S)\frac{\Delta S}{S}$$

$$VaR = \alpha\sigma|\Delta_0|S_0$$

Duration model

$$\Delta P = -DP\Delta y$$

$$\sigma_P = DP\sigma_y$$

5.3 Delta-Normal Methods

Exposure vector for weight vector $\mathbf{w_t}$

$$\mathbf{x_t} = [x_{1,t}, x_{2,t}, ..., x_{N,t}]^T = \mathbf{w_t} V_t$$

$$\sigma^2_{\Delta V} = \sigma^2_p V^2_t = \mathbf{x_t^T C_{t+h} x_t}$$

$$VaR_p = \alpha\sqrt{\mathbf{x_t^T C_{t+h} x_t}}$$

$$\mathbf{C} = \begin{bmatrix} \sigma_1 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2 \end{bmatrix}$$

5.5 Risk Mapping

$$Z_{1.5} = \beta Z_{1.5} + (1 - \beta Z_{1.5}) = Z_1 + Z_2$$

$$y_{1.5} = \gamma y_1 + (1 - \gamma) y_2$$

$$\sigma_{1.5} = \gamma\sigma_1 + (1 - \gamma)\sigma_2$$

γ used to interpolate.

$$a\beta^2 + b\beta + c = 0$$

$$a = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$$

$$b = 2\rho\sigma_1\sigma_2 - 2\sigma_2^2$$

$$c = \sigma_2^2 - \sigma_{1.5}^2$$

$$\mathbf{C} = \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_N \end{bmatrix} corr(X) \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_N \end{bmatrix}$$